# GATE DA 2027

## Probability & Statistics

Topic-wise Question Bank

Weighted by the GATE DA 2024–2026 pattern

**230** questions • **10** topics • **20+** questions per topic

MCQ • MSQ • NAT — exactly the GATE DA question types

Answer key at the end (answers only — no worked solutions)

Statistical tables for the $z$, $t$ and $\chi^2$ distributions are provided in Appendices A–C.
Attempt each topic set in one sitting, then check against the key.

Prepared for Ashutosh • August 2026

## How this book is built

**Why the question counts differ across topics.** Probability & Statistics has been worth **16, 18 and 13 marks** in GATE DA 2024, 2025 and 2026 respectively — an average of about **15–16 marks**, which makes it the single most reliable scoring block in the Engineering Mathematics part of the paper. Within that block the marks are *not* spread evenly. The number of questions given here for each topic follows the share shown in Figure 1.

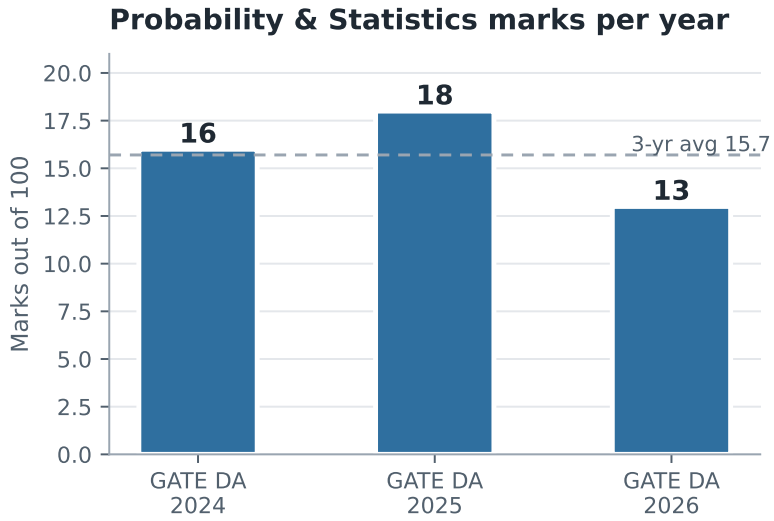



Figure 1(a): Probability & Statistics marks in the last three GATE DA papers.

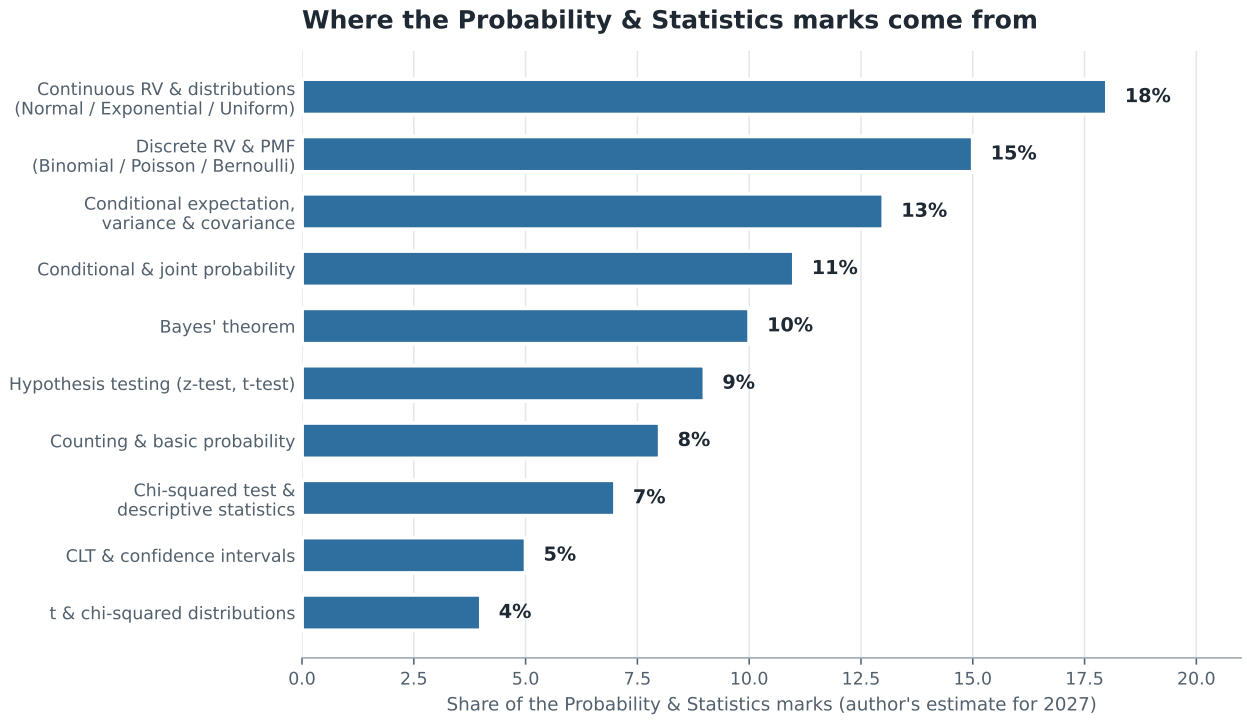



Figure 1(b): Estimated topic-wise share for GATE DA 2027, based on the 2024–2026 papers.

## What the past papers actually reward

- **Distributions dominate.** Normal, Exponential (memorylessness), Poisson, Binomial, Bernoulli and Uniform together account for roughly a third of the block every year. Most of these questions test a *property*, not a long calculation.
- **Bayes' theorem appears almost every year**, usually as a 2-mark NAT with a disease-test, machine-defect or classifier story.
- **Expectation, variance, covariance and correlation** are asked both directly and inside Machine Learning questions — so this topic pays twice.
- **Statistical inference (CLT, confidence intervals, $z/t/\chi^2$ tests)** entered the syllabus explicitly and is now examinable; expect 1–2 questions.
- **Counting** shows up as a single 1-mark question — cheap marks, do not skip it.

## Question types and marking (same as the real paper)

| Type | Description | Negative marking |
|---|---|---|
| MCQ | Four options, exactly one correct | $-1/3$ (1-mark) or $-2/3$ (2-mark) |
| MSQ | Four options, one *or more* correct; no partial credit | None |
| NAT | Numerical answer, entered on a virtual keypad | None |

## How to use it

1. Do one topic set in a single timed sitting: allow **1.5 minutes** per 1-mark question and **3 minutes** per 2-mark question.
2. Do *not* look at the key mid-set. Mark every question you guessed with a dot.
3. Score yourself with GATE negative marking. Anything below 70% on a topic means the concept, not the practice, is the problem — revise before re-attempting.
4. Re-attempt only the wrong and the dotted questions after a week.

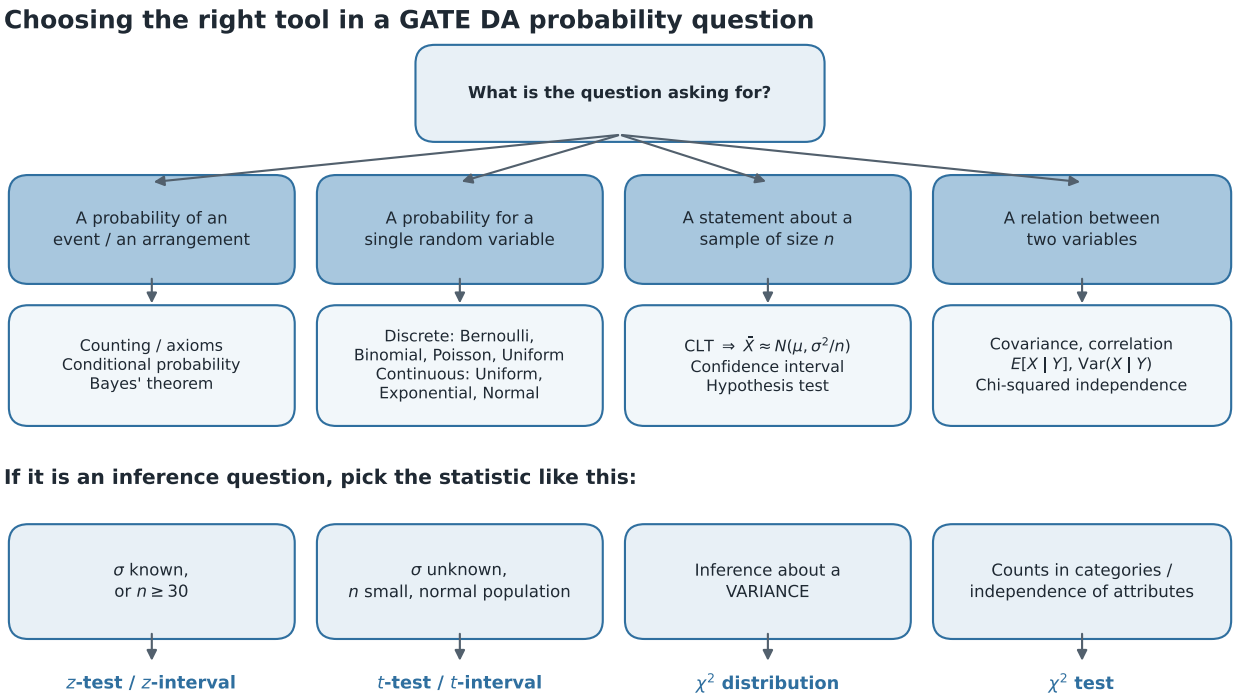



Figure 2: The decision map most 2-mark questions are really testing — name the tool first, then compute.

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# 1. Counting & Basic Probability

**24 questions** • 5 one-mark, 19 two-mark • Suggested time: 64 minutes

**Q.1** [MCQ • 1 Mark • Easy]

A fair coin is tossed 5 times. The probability of obtaining exactly 3 heads is

(A) $\frac{3}{16}$ (B) $\frac{3}{8}$ (C) $\frac{5}{16}$ (D) $\frac{1}{2}$

**Q.2** [NAT • 1 Mark • Easy]

The number of distinct arrangements of the letters of the word **STATISTICS** is _____.

**Q.3** [MCQ • 1 Mark • Easy]

For two independent events $A$ and $B$ with $P(A) = 0.4$ and $P(B) = 0.5$, the value of $P(A \cup B)$ is

(A) 0.70 (B) 0.45 (C) 0.20 (D) 0.90

**Q.4** [NAT • 1 Mark • Easy]

Two fair six-sided dice are rolled once. The probability that the sum of the two faces equals 8 is _____ (rounded off to three decimal places).

**Q.5** [MCQ • 2 Marks • Moderate]

A committee of 5 members is chosen at random from a group of 6 men and 4 women. The probability that the committee contains exactly 3 men is

(A) $\frac{5}{21}$ (B) $\frac{10}{21}$ (C) $\frac{1}{2}$ (D) $\frac{20}{63}$

**Q.6** [NAT • 2 Marks • Moderate]

The number of ways of distributing 10 identical balls into 4 distinct boxes such that no box remains empty is _____.

**Q.7** [MSQ • 2 Marks • Moderate]

Let $A$ and $B$ be arbitrary events in a sample space $S$. Which of the following statement(s) is/are **always** true?

(A) $P(A^c \cap B^c) = 1 - P(A \cup B)$ (B) $P(A \cup B) \leq P(A) + P(B)$

(C) $P(A \cap B) \geq P(A) + P(B) - 1$ (D) If $P(A) = 0$ then $A = \varnothing$

**Q.8** [NAT • 2 Marks • Moderate]

Using the digits $\{1, 2, 3, 4, 5, 6, 7\}$ without repetition, the number of 4-digit numbers that are divisible by 5 is _____.

**Q.9** [MCQ • 2 Marks • Moderate]

Five people are selected at random. Assuming each person is equally likely to be born on any day of the week, the probability that all five were born on different days of the week is closest to

(A) 0.083 (B) 0.300 (C) 0.250 (D) 0.150

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**Q.10** [NAT • 2 Marks • Moderate]

The number of onto (surjective) functions from a set of 5 elements to a set of 3 elements is _____.

---

**Q.11** [MCQ • 2 Marks • Moderate]

A bag contains 5 red and 3 blue balls. Three balls are drawn at random without replacement. The probability that at least one blue ball is drawn is

(A) $\frac{15}{28}$ (B) $\frac{5}{28}$ (C) $\frac{23}{28}$ (D) $\frac{3}{8}$

---

**Q.12** [NAT • 2 Marks • Hard]

The number of non-negative integer solutions of $x_1 + x_2 + x_3 + x_4 = 12$ subject to $x_i \leq 6$ for every $i$ is _____.

---

**Q.13** [MSQ • 2 Marks • Moderate]

Let $A$ and $B$ be events with $P(A) = 0.6$ and $P(B) = 0.3$. Which of the following value(s) is/are **possible**?

(A) $P(A \cap B) = 0.1$ (B) $P(A \cap B) = 0.3$ (C) $P(A \cup B) = 0.5$ (D) $P(A \cup B) = 0.9$

---

**Q.14** [NAT • 2 Marks • Hard]

Eight people are seated at random around a circular table. The probability that two particular persons sit next to each other is _____ (rounded off to three decimal places).

---

**Q.15** [MCQ • 2 Marks • Hard]

The number of binary strings of length 8 that contain no two consecutive 1s is

(A) 34 (B) 89 (C) 128 (D) 55

---

**Q.16** [NAT • 2 Marks • Moderate]

Three fair six-sided dice are rolled once. The probability that all three faces show different numbers is _____ (rounded off to three decimal places).

---

**Q.17** [MCQ • 2 Marks • Moderate]

The letters of the word **ARRANGE** are arranged at random. The probability that the two R's occur together is

(A) $\frac{1}{7}$ (B) $\frac{1}{21}$ (C) $\frac{2}{7}$ (D) $\frac{3}{7}$

---

**Q.18** [NAT • 1 Mark • Moderate]

The number of 5-card hands dealt from a standard 52-card deck that contain exactly two aces is _____.

---

**Q.19** [MCQ • 2 Marks • Moderate]

Given $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$, the probability that exactly one of $A$ and $B$ occurs is

(A) 0.9 (B) 0.7 (C) 0.3 (D) 0.5

---

**Q.20** [NAT • 2 Marks • Moderate]

A fair die is rolled repeatedly until a 6 appears. The probability that more than 4 rolls are required is _____ (rounded off to three decimal places).

---

**Q.21** [MSQ • 2 Marks • Moderate]

Let $A$ and $B$ be independent events with $0 < P(A) < 1$ and $0 < P(B) < 1$. Which of the following statement(s) is/are true?

(A) $A$ and $B$ are mutually exclusive

(B) $A^c$ and $B^c$ are independent

(C) $P(A \mid B) = P(A)$

(D) $A$ and $B^c$ are independent

---

**Q.22** [NAT • 2 Marks • Hard]

The number of ways of arranging 4 boys and 4 girls in a row so that no two boys are adjacent is _____.

---

**Q.23** [MCQ • 2 Marks • Hard]

A permutation of $\{1, 2, 3, 4, 5\}$ is chosen uniformly at random. The probability that no element occupies its own position is

(A) $\frac{3}{10}$ (B) $\frac{44}{125}$ (C) $\frac{1}{3}$ (D) $\frac{11}{30}$

---

**Q.24** [NAT • 2 Marks • Moderate]

Two distinct numbers are chosen at random from $\{1, 2, \ldots, 10\}$. The probability that their sum is even is _____ (rounded off to three decimal places).

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## 2. Conditional & Joint Probability

**24 questions** • 4 one-mark, 20 two-mark • Suggested time: 66 minutes

**Q.1** [MCQ • 1 Mark • Easy]

If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$, then $P(A \mid B)$ equals

(A) 0.40 (B) 0.50 (C) 0.80 (D) 0.25

**Q.2** [NAT • 1 Mark • Easy]

A card is drawn at random from a standard 52-card deck. Given that the card drawn is a face card (J, Q, K), the probability that it is a king is _____ (rounded off to three decimal places).

**Q.3** [MCQ • 1 Mark • Easy]

Two fair dice are rolled. Given that the sum of the faces is 7, the probability that the first die shows a 3 is

(A) $\frac{1}{2}$ (B) $\frac{1}{6}$ (C) $\frac{1}{3}$ (D) $\frac{1}{12}$

**Q.4** [NAT • 1 Mark • Easy]

If $P(A) = 0.3$ and $P(B \mid A) = 0.4$, then $P(A \cap B) =$ _____.

**Q.5** [NAT • 2 Marks • Moderate]

The joint PMF of the discrete random variables $X$ and $Y$ is given below.

| $p(x, y)$ | $Y = 0$ | $Y = 1$ | $Y = 2$ |
|---|---|---|---|
| $X = 0$ | 0.10 | 0.05 | 0.15 |
| $X = 1$ | 0.20 | 0.10 | 0.05 |
| $X = 2$ | 0.05 | 0.20 | 0.10 |

The value of $P(X = 1 \mid Y = 1)$ is _____ (rounded off to three decimal places).

**Q.6** [NAT • 2 Marks • Moderate]

For the joint PMF of the previous question, $E[X] =$ _____ (rounded off to two decimal places).

| $p(x, y)$ | $Y = 0$ | $Y = 1$ | $Y = 2$ |
|---|---|---|---|
| $X = 0$ | 0.10 | 0.05 | 0.15 |
| $X = 1$ | 0.20 | 0.10 | 0.05 |
| $X = 2$ | 0.05 | 0.20 | 0.10 |

**Q.7** [MSQ • 2 Marks • Moderate]

For the joint PMF given in Q.5 of this set, which of the following statement(s) is/are correct?

(A) $P(X = 2) = 0.35$ (B) $X$ and $Y$ are *not* independent

(C) $P(Y = 0) = 0.30$ (D) $P(X \leq 1,\ Y \geq 1) = 0.35$

**Q.8** [MCQ • 2 Marks • Moderate]

An urn contains 3 white and 2 black balls. Two balls are drawn one after the other without replacement. The probability that the *second* ball drawn is white is

(A) $\frac{2}{5}$ (B) $\frac{1}{2}$ (C) $\frac{9}{25}$ (D) $\frac{3}{5}$

**Q.9** [NAT • 2 Marks • Moderate]

Machine $A$ produces 60% of the items of a factory and machine $B$ produces the remaining 40%. The proportion of defective items produced by $A$ and $B$ are 2% and 5% respectively. The probability that a randomly chosen item is defective is _____ (rounded off to three decimal places).

**Q.10** [MCQ • 2 Marks • Moderate]

The joint PDF of $(X, Y)$ is $f(x, y) = c\,xy$ for $0 < x < 1,\ 0 < y < 1$ and 0 otherwise. The value of $c$ is

(A) 2 (B) $\frac{1}{4}$ (C) 4 (D) 1

**Q.11** [NAT • 2 Marks • Moderate]

Let $f(x, y) = 4xy$ for $0 < x < 1,\ 0 < y < 1$ and 0 otherwise. Then $P\left(X < \frac{1}{2},\ Y < \frac{1}{2}\right) =$ _____ (rounded off to four decimal places).

**Q.12** [MCQ • 2 Marks • Moderate]

For the joint PDF $f(x, y) = 4xy$ on the unit square, the value of $P\left(X < \frac{1}{2} \mid Y < \frac{1}{2}\right)$ is

(A) 0.50 (B) 0.125 (C) 0.25 (D) 0.0625

**Q.13** [NAT • 2 Marks • Moderate]

Three cards are drawn at random without replacement from a standard 52-card deck. The probability that all three are hearts is _____ (rounded off to four decimal places).

**Q.14** [MCQ • 2 Marks • Moderate]

If $P(A \mid B) = P(A \mid B^c)$ with $0 < P(B) < 1$, then it must be true that

(A) $P(A) = P(B)$ (B) $A$ and $B$ are independent

(C) $P(B) = 0.5$ (D) $A$ and $B$ are mutually exclusive

**Q.15** [NAT • 2 Marks • Hard]

A box contains 4 good and 2 defective items. Items are tested one at a time at random without replacement. The probability that the second defective item is identified exactly on the third test is _____ (rounded off to three decimal places).

**Q.16** [MSQ • 2 Marks • Moderate]

For events $A$ and $B$ with $0 < P(B) < 1$, which of the following identity/identities is/are always

correct?

(A) $P(A \mid B) + P(A^c \mid B) = 1$ (B) $P(A \mid B) + P(A \mid B^c) = 1$

(C) $P(A \mid B)P(B) = P(B \mid A)P(A)$ (D) If $A \subseteq B$ then $P(A \mid B) = \dfrac{P(A)}{P(B)}$

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**Q.17** [NAT • 2 Marks • Moderate]

$X$ and $Y$ are independent Bernoulli random variables with $P(X = 1) = 0.3$ and $P(Y = 1) = 0.6$. Then $P(X + Y = 1) =$ _____ (rounded off to two decimal places).

---

**Q.18** [MCQ • 2 Marks • Moderate]

For two events, $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A \cup B) = \frac{2}{3}$. Then $P(A \mid B)$ equals

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

---

**Q.19** [NAT • 2 Marks • Hard]

The joint PDF of $(X, Y)$ is $f(x, y) = 2$ for $0 < x < y < 1$ and 0 otherwise. Then $P(Y > 0.5) =$ _____ (rounded off to two decimal places).

---

**Q.20** [MCQ • 2 Marks • Hard]

For the joint PDF $f(x, y) = 2$ on $\{0 < x < y < 1\}$, the conditional expectation $E[X \mid Y = y]$ equals

(A) $\dfrac{1 + y}{2}$ (B) $y$ (C) $\dfrac{1}{2}$ (D) $\dfrac{y}{2}$

---

**Q.21** [NAT • 2 Marks • Moderate]

A student attempts a multiple-choice question having 4 options with exactly one correct answer. With probability 0.7 the student knows the answer and answers correctly; otherwise the student guesses uniformly at random. The probability that the student answers the question correctly is _____ (rounded off to three decimal places).

---

**Q.22** [MSQ • 2 Marks • Hard]

Let $X$ and $Y$ be two random variables. Which of the following condition(s) **necessarily** implies that $X$ and $Y$ are independent?

(A) $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ for all $x, y$ (B) $F_{X,Y}(x, y) = F_X(x) F_Y(y)$ for all $x, y$

(C) $\mathrm{Cov}(X, Y) = 0$ (D) $E[XY] = E[X]\,E[Y]$

---

**Q.23** [NAT • 2 Marks • Hard]

Three fair coins are tossed. Given that at least one head appeared, the probability that all three are heads is _____ (rounded off to three decimal places).

---

**Q.24** [MCQ • 2 Marks • Moderate]

$A$ and $B$ are two events such that $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \mid B) = 0.4$. The value of $P(B \mid A)$ is

(A) 0.50 (B) 0.20 (C) 0.32 (D) 0.80

## 3. Bayes' Theorem

**22 questions** • 3 one-mark, 19 two-mark • Suggested time: 61 minutes

**Q.1** [MCQ • 1 Mark • Easy]

Box 1 contains 2 red and 3 blue balls; Box 2 contains 4 red and 1 blue ball. A box is chosen uniformly at random and one ball is drawn from it. Given that the drawn ball is red, the probability that it came from Box 1 is

(A) $\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{1}{5}$ (D) $\frac{1}{2}$

**Q.2** [NAT • 1 Mark • Easy]

A disease affects 1% of a population. A diagnostic test has sensitivity 99% and specificity 95%. If a randomly chosen person tests positive, the probability that the person actually has the disease is _____ (rounded off to three decimal places).

**Q.3** [MCQ • 1 Mark • Easy]

In Bayes' theorem $P(A_i \mid B) = \dfrac{P(B \mid A_i)P(A_i)}{\sum_j P(B \mid A_j)P(A_j)}$, the denominator is obtained using

(A) the multiplication rule for independent events (B) the law of total probability

(C) the inclusion–exclusion principle (D) Chebyshev's inequality

**Q.4** [NAT • 2 Marks • Moderate]

Machine $A$ makes 60% of a factory's items with a 2% defect rate; machine $B$ makes the remaining 40% with a 5% defect rate. Given that a randomly selected item is defective, the probability that it was made by machine $B$ is _____ (rounded off to three decimal places).

**Q.5** [MCQ • 2 Marks • Moderate]

Machines $M_1, M_2, M_3$ produce 50%, 30% and 20% of a factory's output with defect rates 3%, 4% and 5% respectively. Given that an item is defective, the probability that it came from $M_3$ is closest to

(A) 0.270 (B) 0.200 (C) 0.324 (D) 0.135

**Q.6** [NAT • 2 Marks • Moderate]

For the three-machine setting of the previous question, the probability that a randomly selected item is defective is _____ (rounded off to three decimal places).

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about Bayes' theorem is/are correct?

(A) The posterior probability is always greater than the prior probability

(B) The events $A_1, \ldots, A_n$ must form a partition of the sample space

(C) Posterior $\propto$ likelihood $\times$ prior

(D) For a partition $\{A_i\}$, $\sum_i P(A_i \mid B) = 1$

**Q.8** [NAT • 2 Marks • Moderate]

A box contains three coins: two are fair and one has heads on both faces. A coin is picked at random and tossed once, showing a head. The probability that the two-headed coin was picked is _____ (rounded off to two decimal places).

**Q.9** [MCQ • 2 Marks • Moderate]

A box contains two fair coins and one two-headed coin. A coin is picked at random and tossed *twice*; both tosses show heads. The probability that the two-headed coin was picked is

(A) $\frac{4}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{2}$ (D) $\frac{1}{3}$

**Q.10** [NAT • 2 Marks • Moderate]

In an e-mail corpus 40% of messages are spam. The word "free" occurs in 20% of spam messages and in 2% of non-spam messages. Given that a message contains "free", the probability that it is spam is _____ (rounded off to three decimal places).

**Q.11** [MCQ • 2 Marks • Moderate]

The "naive" assumption in a Naive Bayes classifier is that

(A) the features are marginally independent of each other

(B) the features are conditionally independent given the class label

(C) the class prior is uniform over all classes

(D) the features follow a Gaussian distribution

**Q.12** [NAT • 2 Marks • Moderate]

In the Monty Hall problem with 3 doors (one car, two goats), the contestant picks a door, the host — who knows the contents — opens a different door revealing a goat, and offers a switch. The probability of winning the car if the contestant switches is _____ (rounded off to three decimal places).

**Q.13** [MCQ • 2 Marks • Moderate]

A screening test for a condition of prevalence 0.1% has sensitivity 100% and a false-positive rate of 1%. The probability that a person who tests positive actually has the condition is closest to

(A) 0.091 (B) 0.010 (C) 0.500 (D) 0.909

**Q.14** [NAT • 2 Marks • Moderate]

Bag $A$ contains 5 red and 5 green balls; bag $B$ contains 3 red and 7 green balls. Bag $A$ is chosen with probability 0.6 and bag $B$ with probability 0.4; one ball is then drawn. Given that the ball is green, the probability that bag $A$ was chosen is _____ (rounded off to three decimal places).

**Q.15** [MSQ • 2 Marks • Moderate]

In a screening problem let $p$ be the prevalence, $s$ the sensitivity and $t$ the specificity of the test. Keeping the other two quantities fixed, which of the following change(s) will **increase** the posterior probability $P(\text{disease} \mid \text{positive})$?

(A) Increasing $t$ (B) Increasing $p$

(C) Increasing the false-positive rate $1 - t$ (D) Increasing $s$

---

**Q.16** [NAT • 2 Marks • Hard]

In a city 85% of the cabs are green and 15% are blue. A witness identifies the colour of a cab correctly 80% of the time. The witness reports that a cab involved in an incident was blue. The probability that the cab was actually blue is _____ (rounded off to three decimal places).

---

**Q.17** [MCQ • 2 Marks • Hard]

A coin has bias $\theta$ where the prior is $P(\theta = 0.5) = P(\theta = 0.8) = 0.5$. The coin is tossed three times and all three tosses give heads. The posterior probability $P(\theta = 0.8 \mid 3 \text{ heads})$ is closest to

(A) 0.640 (B) 0.512 (C) 0.196 (D) 0.804

---

**Q.18** [NAT • 2 Marks • Hard]

Box 1 contains 6 red and 4 white balls; Box 2 contains 4 red and 6 white balls. One ball is transferred at random from Box 1 to Box 2 and then a ball is drawn at random from Box 2. Given that the ball drawn is red, the probability that the transferred ball was red is _____ (rounded off to three decimal places).

---

**Q.19** [MCQ • 2 Marks • Moderate]

If $P(A \mid B) = P(A)$ with $P(A) > 0$ and $P(B) > 0$, then $P(B \mid A)$ equals

(A) 1 (B) $P(A)P(B)$ (C) $P(B)$ (D) $P(A)$

---

**Q.20** [NAT • 2 Marks • Moderate]

Over a binary channel the symbols 0 and 1 are transmitted with probabilities 0.6 and 0.4 respectively, and the channel flips a transmitted bit with probability 0.1. Given that a 1 was received, the probability that a 1 was transmitted is _____ (rounded off to three decimal places).

---

**Q.21** [MSQ • 2 Marks • Hard]

For the binary channel described in the previous question, which of the following statement(s) is/are correct?

(A) $P(\text{receive } 0) = 0.58$ (B) $P(0 \text{ sent} \mid 1 \text{ received}) = \dfrac{1}{7}$

(C) $P(\text{receive } 1) = 0.42$ (D) $P(1 \text{ sent} \mid 0 \text{ received}) = 0.40$

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**Q.22** [NAT • 2 Marks • Moderate]

A disease has prevalence 1%. Two *conditionally independent* tests are applied, each with sensitivity 90% and false-positive rate 10%. Given that both tests are positive, the probability that the person

has the disease is _____ (rounded off to two decimal places).

## 4. Discrete Random Variables & PMF

**26 questions** • 4 one-mark, 22 two-mark • Suggested time: 72 minutes

**Q.1** [MCQ • 1 Mark • Easy]

If $X \sim \text{Binomial}(n = 10,\ p = 0.3)$, then $\text{Var}(X)$ equals

(A) 0.21 (B) 2.1 (C) 3.0 (D) 1.45

---

**Q.2** [NAT • 1 Mark • Easy]

If $X \sim \text{Poisson}(\lambda = 2)$, then $P(X = 0) =$ _____ (rounded off to three decimal places).

---

**Q.3** [MCQ • 1 Mark • Easy]

For a Bernoulli random variable $X$ with success probability $p$, the variance of $X$ is

(A) $p$ (B) $p^2$ (C) $p(1-p)$ (D) $\sqrt{p(1-p)}$

---

**Q.4** [NAT • 1 Mark • Easy]

If $X \sim \text{Binomial}(n = 5,\ p = 0.4)$, then $P(X = 2) =$ _____ (rounded off to three decimal places).

---

**Q.5** [MCQ • 2 Marks • Moderate]

A random variable $X$ is uniformly distributed on $\{1, 2, 3, 4, 5, 6\}$. Then $\text{Var}(X)$ equals

(A) $\frac{7}{2}$ (B) $\frac{91}{6}$ (C) $\frac{35}{6}$ (D) $\frac{35}{12}$

---

**Q.6** [NAT • 2 Marks • Moderate]

If $X \sim \text{Poisson}(\lambda = 3)$, then $P(X \geq 2) =$ _____ (rounded off to three decimal places).

---

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the Binomial$(n, p)$ distribution is/are correct?

(A) It is the distribution of the sum of $n$ i.i.d. Bernoulli$(p)$ variables

(B) As $n \to \infty$, $p \to 0$ with $np = \lambda$ fixed, it converges to Poisson$(\lambda)$

(C) Its variance is $np$

(D) Its mean is $np$

---

**Q.8** [NAT • 2 Marks • Moderate]

A fair six-sided die is rolled repeatedly until a 6 appears. The expected number of rolls required is _____.

---

**Q.9** [MCQ • 2 Marks • Moderate]

Let $X \sim \text{Geometric}(p)$ taking values in $\{1, 2, 3, \dots\}$. Then $P(X > k)$ for a positive integer $k$ equals

(A) $(1-p)^k$ (B) $1-(1-p)^k$ (C) $(1-p)^{k-1}$ (D) $p(1-p)^k$

---

**Q.10** [NAT • 2 Marks • Moderate]

The number of accidents at a junction follows a Poisson distribution with mean 2 per week. The probability of exactly 3 accidents in a period of two weeks is _____ (rounded off to three decimal places).

---

**Q.11** [MCQ • 2 Marks • Moderate]

If $X \sim \text{Binomial}(n, p)$ with $E[X] = 6$ and $\text{Var}(X) = 4.2$, then $n$ equals

(A) 15 (B) 30 (C) 20 (D) 10

---

**Q.12** [NAT • 2 Marks • Moderate]

The mode of the Binomial$(n = 10,\ p = 0.35)$ distribution is _____.

---

**Q.13** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the Poisson distribution is/are correct?

(A) The sum of two independent Poisson variables is Poisson

(B) Its moment generating function is $e^{\lambda(e^t-1)}$

(C) Its mean equals its variance

(D) It possesses the memoryless property

---

**Q.14** [NAT • 2 Marks • Hard]

If $X \sim \text{Poisson}(\lambda = 4)$, then $E[X^2] =$ _____.

---

**Q.15** [MCQ • 2 Marks • Moderate]

A fair coin is tossed 4 times. The probability of getting at least 3 heads is

(A) $\frac{3}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{16}$ (D) $\frac{11}{16}$

---

**Q.16** [NAT • 2 Marks • Moderate]

If $X \sim \text{Binomial}(n = 8,\ p = 0.5)$, then $P(X = 4) =$ _____ (rounded off to three decimal places).

---

**Q.17** [MCQ • 2 Marks • Hard]

A discrete random variable $X$ has PMF $P(X = k) = \dfrac{c}{2^k}$ for $k = 1, 2, 3, \ldots$. Then $E[X]$ equals

(A) 3 (B) $\infty$ (C) 2 (D) 1

---

**Q.18** [NAT • 2 Marks • Hard]

A fair coin is tossed repeatedly until two consecutive heads appear. The expected number of tosses

required is _____.

---

**Q.19** [MCQ • 2 Marks • Moderate]

If $X \sim \text{Poisson}(\lambda)$ and $P(X = 1) = P(X = 2)$, then $\lambda$ equals

(A) 3 (B) 1 (C) $\frac{1}{2}$ (D) 2

---

**Q.20** [NAT • 2 Marks • Moderate]

A biased coin with $P(\text{head}) = 0.25$ is tossed 12 times. The variance of the number of heads obtained is _____ (rounded off to two decimal places).

---

**Q.21** [MSQ • 2 Marks • Hard]

Let $X$ be uniformly distributed on $\{1, 2, \ldots, n\}$. Which of the following statement(s) is/are correct?

(A) $E[X] = \dfrac{n+1}{2}$

(B) $P(X = k) = \dfrac{1}{n}$ for $k = 1, \ldots, n$

(C) $\text{Var}(X) = \dfrac{n^2}{12}$

(D) $\text{Var}(X) = \dfrac{n^2 - 1}{12}$

---

**Q.22** [NAT • 2 Marks • Moderate]

An inspector examines 20 independently chosen items from a production line in which 5% of items are defective. The probability that at most one defective item is found is _____ (rounded off to three decimal places).

---

**Q.23** [MCQ • 2 Marks • Hard]

A fair coin is tossed repeatedly. The probability that the third head occurs exactly on the fifth toss is

(A) $\frac{1}{8}$ (B) $\frac{5}{16}$ (C) $\frac{1}{16}$ (D) $\frac{3}{16}$

---

**Q.24** [NAT • 2 Marks • Moderate]

$X$ and $Y$ are independent Poisson random variables with means 2 and 3 respectively. Then $P(X+Y = 1) =$ _____ (rounded off to three decimal places).

---

**Q.25** [MCQ • 2 Marks • Moderate]

The moment generating function of $X \sim \text{Binomial}(n, p)$ is

(A) $\dfrac{pe^t}{1 - (1-p)e^t}$

(B) $\left(1 - p + pe^t\right)^n$

(C) $\left(pe^t\right)^n$

(D) $e^{np(e^t - 1)}$

---

**Q.26** [NAT • 2 Marks • Hard]

A discrete random variable $X$ has PMF $P(X = k) = \dfrac{k}{15}$ for $k = 1, 2, 3, 4, 5$. Then $E[X] =$ _____ (rounded off to three decimal places).

## 5. Continuous Random Variables & PDF/CDF

**26 questions** • 4 one-mark, 22 two-mark • Suggested time: 72 minutes

**Q.1** [MCQ • 1 Mark • Easy]

If $X \sim \text{Uniform}(2, 8)$, then $\text{Var}(X)$ equals

(A) 5 (B) 1 (C) 3 (D) 36

**Q.2** [NAT • 1 Mark • Easy]

If $X$ is exponential with rate $\lambda = 0.5$, then $P(X > 4) =$ _____ (rounded off to three decimal places).

**Q.3** [MCQ • 1 Mark • Easy]

A function $f$ is a valid probability density function on $\mathbb{R}$ if and only if

(A) $f(x) \geq 0$ for all $x$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(B) $f$ is increasing and bounded by 1

(C) $f$ is continuous and $\int_{-\infty}^{\infty} f(x)\,dx = 1$

(D) $0 \leq f(x) \leq 1$ for all $x$

**Q.4** [NAT • 1 Mark • Easy]

If $Z \sim N(0, 1)$, then $P(Z < 1.5) =$ _____ (rounded off to three decimal places). *(Use the standard normal table in Appendix A.)*

**Q.5** [MCQ • 2 Marks • Moderate]

If $X \sim N(50,\ 10^2)$, then $P(40 < X < 60)$ is closest to

(A) 0.3174 (B) 0.6826 (C) 0.5000 (D) 0.9544

**Q.6** [NAT • 2 Marks • Moderate]

The lifetime of a component is exponentially distributed with mean 5 years. Given that it has already survived 2 years, the probability that it survives beyond 3 years is _____ (rounded off to three decimal places).

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the exponential distribution with rate $\lambda$ is/are correct?

(A) Its mode is $1/\lambda$ (B) Its variance is $1/\lambda^2$

(C) Its mean is $1/\lambda$ (D) It is memoryless

**Q.8** [NAT • 2 Marks • Moderate]

A random variable has PDF $f(x) = kx^2$ for $0 \leq x \leq 3$ and 0 otherwise. The value of $k$ is _____

(rounded off to three decimal places).

---

**Q.9** [MCQ • 2 Marks • Moderate]

For the PDF $f(x) = \frac{1}{9}x^2$ on $[0, 3]$, the value of $E[X]$ is

(A) 2.00 (B) 1.50 (C) 3.00 (D) 2.25

---

**Q.10** [NAT • 2 Marks • Moderate]

A continuous random variable has CDF $F(x) = 1 - e^{-x/3}$ for $x > 0$. Its median is _____ (rounded off to three decimal places).

---

**Q.11** [MCQ • 2 Marks • Moderate]

Let $U \sim \text{Uniform}(0, 1)$ and $Y = -2 \ln U$. The distribution of $Y$ is

(A) standard normal (B) exponential with mean 2

(C) exponential with mean $\frac{1}{2}$ (D) uniform on $(0, 2)$

---

**Q.12** [NAT • 2 Marks • Moderate]

IQ scores are modelled as $N(100, 15^2)$. The probability that a randomly chosen person has an IQ above 130 is _____ (rounded off to three decimal places).

---

**Q.13** [MCQ • 2 Marks • Moderate]

In a data set with mean 70 and standard deviation 12, the $z$-score of the observation 85 is

(A) 1.25 (B) 1.50 (C) 15.00 (D) 0.80

---

**Q.14** [NAT • 2 Marks • Hard]

For $X \sim N(\mu, \sigma^2)$ it is given that $P(X < 70) = 0.9772$ and $P(X < 60) = 0.8413$. The value of $\sigma$ is _____.

---

**Q.15** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the normal distribution $N(\mu, \sigma^2)$ is/are correct?

(A) Mean, median and mode coincide (B) The density is symmetric about $\mu$

(C) $P(X = \mu) = \dfrac{1}{\sigma\sqrt{2\pi}}$ (D) The total area under the density curve is 1

---

**Q.16** [NAT • 2 Marks • Moderate]

If $X \sim \text{Uniform}(0, 10)$, then $P(X > 7 \mid X > 4) =$ _____ (rounded off to two decimal places).

---

**Q.17** [MCQ • 2 Marks • Hard]

If $X$ is exponential, the identity $P(X > s + t \mid X > s) = P(X > t)$ for all $s, t > 0$ is known as

(A) the memoryless property
(B) the inversion property
(C) the reproductive property
(D) the law of total expectation

**Q.18** [NAT • 2 Marks • Moderate]

A random variable has PDF $f(x) = 2e^{-2x}$ for $x > 0$. Its variance is _____ (rounded off to two decimal places).

**Q.19** [MCQ • 2 Marks • Moderate]

For a continuous random variable $X$ with CDF $F$, the probability $P(a < X \leq b)$ equals

(A) $\dfrac{F(b)}{F(a)}$ (B) $f(b) - f(a)$ (C) $F(a) - F(b)$ (D) $F(b) - F(a)$

**Q.20** [NAT • 2 Marks • Hard]

If $Z \sim N(0, 1)$, then $P(|Z| > 1.96) =$ _____ (rounded off to two decimal places).

**Q.21** [MSQ • 2 Marks • Hard]

Let $X \sim \text{Uniform}(a, b)$ with $a < b$. Which of the following statement(s) is/are correct?

(A) $F(x) = \dfrac{x - a}{(b - a)^2}$ for $a \leq x \leq b$
(B) $\text{Var}(X) = \dfrac{(b - a)^2}{12}$
(C) $E[X] = \dfrac{a + b}{2}$
(D) $f(x) = \dfrac{1}{b - a}$ for $a \leq x \leq b$

**Q.22** [NAT • 2 Marks • Moderate]

Inter-arrival times at a service counter are exponential with mean 4 minutes. The probability that the next arrival occurs within 2 minutes is _____ (rounded off to three decimal places).

**Q.23** [MCQ • 2 Marks • Hard]

A random variable has PDF $f(x) = 3x^2$ for $0 < x < 1$. Then $P(X > 0.5)$ equals

(A) 0.125 (B) 0.750 (C) 0.875 (D) 0.500

**Q.24** [NAT • 2 Marks • Hard]

If $X \sim N(20,\ 4^2)$, the value of $c$ satisfying $P(X > c) = 0.05$ is _____ (rounded off to two decimal places).

**Q.25** [MCQ • 2 Marks • Moderate]

If $Z \sim N(0, 1)$, then $Z^2$ follows

(A) a standard normal distribution
(B) a chi-squared distribution with 1 degree of freedom
(C) a $t$-distribution with 1 degree of freedom
(D) an exponential distribution with rate 1

**Q.26** [NAT • 2 Marks • Hard]

Let $X \sim \text{Uniform}(0, 1)$ and $Y = X^2$. The value of the density $f_Y(y)$ at $y = 0.25$ is _____ (rounded off to two decimal places).

## 6. $t$-distribution & Chi-Squared Distribution

**20 questions** • 4 one-mark, 16 two-mark • Suggested time: 54 minutes

**Q.1** [MCQ • 1 Mark • Easy]

Compared with the standard normal distribution, the Student's $t$-distribution with $\nu$ degrees of freedom

(A) has the same tails but a different mean

(B) has lighter tails and approaches $N(0,1)$ as $\nu \to \infty$

(C) has heavier tails and approaches $N(0,1)$ as $\nu \to \infty$

(D) is right-skewed for every finite $\nu$

---

**Q.2** [NAT • 1 Mark • Easy]

If $X \sim \chi^2_8$, then $\mathrm{Var}(X) =$ _____.

---

**Q.3** [MCQ • 1 Mark • Easy]

If $Z_1, Z_2, \ldots, Z_n$ are i.i.d. $N(0,1)$, then $\sum_{i=1}^{n} Z_i^2$ follows

(A) $N(0,n)$ (B) $t_n$ (C) $\chi^2_n$ (D) $\chi^2_{n-1}$

---

**Q.4** [NAT • 1 Mark • Easy]

The variance of a $t$-distribution with 10 degrees of freedom is _____ (rounded off to two decimal places).

---

**Q.5** [MCQ • 2 Marks • Moderate]

The Student's $t$ statistic with $\nu$ degrees of freedom is defined as

(A) $Z^2/\nu$ where $Z \sim N(0,1)$

(B) $\dfrac{Z}{\sqrt{V/\nu}}$ where $Z \sim N(0,1)$, $V \sim \chi^2_\nu$ are independent

(C) $\dfrac{V/\nu}{Z}$ where $Z \sim N(0,1)$, $V \sim \chi^2_\nu$

(D) $\dfrac{Z}{\sqrt{V}}$ where $Z \sim N(0,1)$, $V \sim \chi^2_\nu$

---

**Q.6** [NAT • 2 Marks • Moderate]

A random sample of size 16 is drawn from a normal population. The statistic $\dfrac{(n-1)S^2}{\sigma^2}$ follows a chi-squared distribution with _____ degrees of freedom.

---

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the $\chi^2_k$ distribution is/are correct?

(A) The sum of two independent chi-squared variables with $k_1$ and $k_2$ degrees of freedom is $\chi^2_{k_1+k_2}$

(B) It takes only non-negative values

(C) It is symmetric about 0

(D) It is right-skewed, and the skewness decreases as $k$ increases

---

**Q.8** [NAT • 2 Marks • Moderate]

From the $t$-table, the upper 5% critical value $t_{0.05,\,12}$ is _____ (rounded off to three decimal places). *(Use Appendix B.)*

---

**Q.9** [MCQ • 2 Marks • Moderate]

The chi-squared distribution with 2 degrees of freedom is identical to

(A) the standard normal distribution (B) the exponential distribution with mean 2

(C) the exponential distribution with mean $\frac{1}{2}$ (D) the uniform distribution on $(0, 2)$

---

**Q.10** [NAT • 2 Marks • Moderate]

From the chi-squared table, the upper 5% critical value $\chi^2_{0.05,\,10}$ is _____ (rounded off to three decimal places). *(Use Appendix C.)*

---

**Q.11** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the $t$-distribution with $\nu$ degrees of freedom is/are correct?

(A) It converges in distribution to $N(0, 1)$ as $\nu \to \infty$

(B) Its mean is 0 for $\nu > 1$

(C) It is symmetric about 0

(D) Its variance equals 1 for every $\nu$

---

**Q.12** [NAT • 2 Marks • Moderate]

A random sample of size 25 from a normal population with $\sigma = 5$ gives a sample standard deviation $s = 6$. The value of the statistic $\dfrac{(n-1)s^2}{\sigma^2}$ is _____ (rounded off to two decimal places).

---

**Q.13** [MCQ • 2 Marks • Moderate]

For constructing a confidence interval for the *variance* of a normal population, the appropriate sampling distribution is

(A) uniform (B) Student's $t$ (C) standard normal (D) chi-squared

---

**Q.14** [NAT • 2 Marks • Hard]

A chi-squared random variable has variance 30. Its number of degrees of freedom is _____.

---

**Q.15** [MCQ • 2 Marks • Moderate]

As the number of degrees of freedom $k$ increases, the $\chi^2_k$ distribution

(A) approaches the $t$-distribution

(B) becomes more symmetric and approaches a normal shape

(C) becomes more right-skewed

(D) converges to a point mass at 0

---

**Q.16** [NAT • 2 Marks • Moderate]

From the $t$-table, the value $t_{0.025,\,20}$ is _____ (rounded off to three decimal places). *(Use Appendix B.)*

---

**Q.17** [MCQ • 2 Marks • Hard]

Let $X_1, \ldots, X_{10}$ be i.i.d. $N(0, \sigma^2)$. Then $E\left[\sum_{i=1}^{10} X_i^2\right]$ equals

(A) $\sigma^2$ (B) $20\sigma^2$ (C) $100\sigma^2$ (D) $10\sigma^2$

---

**Q.18** [NAT • 2 Marks • Hard]

The mode of the $\chi^2_9$ distribution is _____.

---

**Q.19** [MCQ • 2 Marks • Moderate]

The two-sided 95% critical value of the $t$-distribution with 9 degrees of freedom is

(A) 2.262 (B) 2.821 (C) 1.960 (D) 1.833

---

**Q.20** [NAT • 2 Marks • Hard]

If $X \sim \chi^2_4$, then $P(X > 9.488) =$ _____ (rounded off to two decimal places). *(Use Appendix C.)*

---

## 7. Central Limit Theorem & Confidence Intervals

**22 questions** • 4 one-mark, 18 two-mark • Suggested time: 60 minutes

**Q.1** [MCQ • 1 Mark • Easy]

The Central Limit Theorem states that, for i.i.d. observations with finite variance,

(A) the sample variance approaches 0 as $n \to \infty$ for every population

(B) the sample mean equals the population mean for large $n$

(C) the distribution of the standardised sample mean approaches $N(0,1)$ as $n \to \infty$, whatever the population distribution

(D) the population distribution becomes normal as $n \to \infty$

---

**Q.2** [NAT • 1 Mark • Easy]

A population has standard deviation $\sigma = 20$. For a random sample of size $n = 100$, the standard error of the sample mean is _____.

---

**Q.3** [MCQ • 1 Mark • Easy]

For a 95% confidence interval for a population mean with $\sigma$ known, the appropriate multiplier is

(A) 2.576 (B) 2.326 (C) 1.960 (D) 1.645

---

**Q.4** [NAT • 1 Mark • Easy]

A sample of size 64 from a population with $\sigma = 16$ gives $\bar{x} = 50$. The margin of error of the 95% confidence interval for $\mu$ is _____ (rounded off to two decimal places).

---

**Q.5** [MCQ • 2 Marks • Moderate]

A random sample of size 36 is drawn from a population with $\mu = 10$ and $\sigma = 6$. Using the CLT, $P(\bar{X} > 11)$ is closest to

(A) 0.0228 (B) 0.1587 (C) 0.3413 (D) 0.8413

---

**Q.6** [NAT • 2 Marks • Moderate]

The minimum sample size required so that the 95% confidence interval for $\mu$ has margin of error at most 2, when $\sigma = 10$, is _____.

---

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the Central Limit Theorem is/are correct?

(A) It requires the population distribution to be normal

(B) It applies to the sample sum as well as to the sample mean

(C) It requires the population variance to be finite

(D) The standard error of the sample mean decreases as $1/\sqrt{n}$

**Q.8** [NAT • 2 Marks • Moderate]

A random sample of size 25 from a normal population gives $\bar{x} = 80$ and $s = 10$. The margin of error of the 95% confidence interval for $\mu$ is _____ (rounded off to three decimal places). *(Use $t_{0.025,\,24} = 2.064$.)*

**Q.9** [MCQ • 2 Marks • Moderate]

The correct interpretation of a 95% confidence interval for $\mu$ is:

(A) 95% of the sample observations lie inside the interval

(B) the sample mean lies in the interval with probability 0.95

(C) the probability that $\mu$ lies in this particular interval is 0.95

(D) if the sampling procedure is repeated many times, about 95% of the intervals so constructed will contain $\mu$

**Q.10** [NAT • 2 Marks • Moderate]

For a sample of size 36 with $\bar{x} = 100$ from a population with $\sigma = 15$, the lower limit of the 99% confidence interval for $\mu$ is _____ (rounded off to two decimal places). *(Use $z_{0.005} = 2.576$.)*

**Q.11** [MCQ • 2 Marks • Moderate]

Keeping the sample size and the data fixed, raising the confidence level from 90% to 99% makes the confidence interval

(A) narrower (B) narrower only if $n > 30$ (C) unchanged in width (D) wider

**Q.12** [NAT • 2 Marks • Hard]

A population is uniform on $(0, 1)$. For a random sample of size 48, the standard deviation of the sample mean is _____ (rounded off to three decimal places).

**Q.13** [MSQ • 2 Marks • Moderate]

The width of a confidence interval for a population mean depends on which of the following?

(A) the observed value of the sample mean (B) the sample size

(C) the population standard deviation (D) the confidence level

**Q.14** [NAT • 2 Marks • Moderate]

A fair coin is tossed 100 times. Using the normal approximation (without continuity correction), the probability of obtaining more than 60 heads is _____ (rounded off to three decimal places).

**Q.15** [MCQ • 2 Marks • Hard]

In a sample of 400 voters, 25% support a proposal. The margin of error of the 95% confidence interval for the population proportion is closest to

(A) 0.0849 (B) 0.0250 (C) 0.0424 (D) 0.0212

**Q.16** [NAT • 2 Marks • Moderate]

A random sample of size 100 is drawn from a population with mean 50 and standard deviation 10. Then $P(48 < \bar{X} < 52) =$ _____ (rounded off to three decimal places).

**Q.17** [MCQ • 2 Marks • Moderate]

With $\sigma$ known and the confidence level unchanged, halving the width of a confidence interval for $\mu$ requires the sample size to be

(A) multiplied by $\sqrt{2}$ (B) divided by 2 (C) multiplied by 2 (D) multiplied by 4

**Q.18** [NAT • 2 Marks • Hard]

Let $S = \sum_{i=1}^{100} X_i$ where the $X_i$ are i.i.d. with mean 2 and variance 4. Using the CLT, $P(S > 220) =$ _____ (rounded off to three decimal places).

**Q.19** [MSQ • 2 Marks • Hard]

Regarding the choice between a $z$-interval and a $t$-interval for a population mean, which statement(s) is/are correct?

(A) For large $n$ the two intervals nearly coincide

(B) For the same data and confidence level, the $t$-interval is always narrower than the $z$-interval

(C) The $t$-interval is used when $\sigma$ is unknown and the population is (approximately) normal

(D) The $z$-interval is used when $\sigma$ is known

**Q.20** [NAT • 2 Marks • Moderate]

A sample of size 49 from a population with $\sigma = 14$ gives $\bar{x} = 35$. The upper limit of the 90% confidence interval for $\mu$ is _____ (rounded off to two decimal places). *(Use $z_{0.05} = 1.645$.)*

**Q.21** [MCQ • 2 Marks • Moderate]

The standard error of the sample mean $\bar{X}$ based on $n$ i.i.d. observations from a population with standard deviation $\sigma$ is

(A) $\dfrac{\sigma^2}{n}$ (B) $\sigma\sqrt{n}$ (C) $\dfrac{\sigma}{n}$ (D) $\dfrac{\sigma}{\sqrt{n}}$

**Q.22** [NAT • 2 Marks • Hard]

For $\sigma = 4$, the minimum sample size required so that $P(|\bar{X} - \mu| \le 0.5) \ge 0.95$ is _____.

## 8. Hypothesis Testing: $z$-test & $t$-test

**22 questions** • 4 one-mark, 18 two-mark • Suggested time: 60 minutes

**Q.1** [MCQ • 1 Mark • Easy]

A Type I error in hypothesis testing is committed when

(A) $H_0$ is rejected although it is true

(B) $H_0$ is not rejected although it is false

(C) the sample size is too small

(D) $H_1$ is rejected although it is true

---

**Q.2** [NAT • 1 Mark • Easy]

For a two-tailed $z$-test at $\alpha = 0.05$, the magnitude of the critical value is _____ (rounded off to two decimal places).

---

**Q.3** [MCQ • 1 Mark • Easy]

The $p$-value of a test is

(A) the probability that $H_0$ is true given the data

(B) the probability, computed assuming $H_0$ is true, of obtaining a test statistic at least as extreme as the observed one

(C) the significance level of the test

(D) the probability of committing a Type II error

---

**Q.4** [NAT • 1 Mark • Easy]

If the probability of a Type II error of a test is $\beta = 0.2$, the power of the test is _____ (rounded off to two decimal places).

---

**Q.5** [MCQ • 2 Marks • Moderate]

A sample of size 64 gives $\bar{x} = 52$. Testing $H_0 : \mu = 50$ against $H_1 : \mu \neq 50$ with $\sigma = 8$, the $p$-value is closest to

(A) 0.9772 (B) 0.0228 (C) 0.0456 (D) 0.0500

---

**Q.6** [NAT • 2 Marks • Moderate]

For the test described in the previous question, the value of the $z$ test statistic is _____ (rounded off to two decimal places).

---

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about hypothesis testing is/are correct?

(A) $H_0$ is rejected when the $p$-value is smaller than $\alpha$

(B) Reducing $\alpha$, with everything else fixed, reduces the power of the test

(C) Increasing the sample size, with everything else fixed, increases the power

(D) The $p$-value is the probability that $H_0$ is true

**Q.8** [NAT • 2 Marks • Moderate]

A random sample of size 16 from a normal population gives $\bar{x} = 105$ and $s = 8$. For testing $H_0 : \mu = 100$, the value of the $t$ statistic is _____ (rounded off to two decimal places).

**Q.9** [MCQ • 2 Marks • Moderate]

For the $t$-test of the previous question against $H_1 : \mu > 100$ at $\alpha = 0.05$ (with $t_{0.05,\,15} = 1.753$), the correct decision is

(A) reject $H_0$, since $2.50 < 1.753$

(B) reject $H_0$, since $2.50 > 1.753$

(C) do not reject $H_0$, since $2.50 > 1.753$

(D) the test is inconclusive because $n < 30$

**Q.10** [NAT • 2 Marks • Moderate]

For a two-tailed one-sample $t$-test with $n = 20$ at $\alpha = 0.01$, the magnitude of the critical value is _____ (rounded off to three decimal places). *(Use Appendix B.)*

**Q.11** [MCQ • 2 Marks • Moderate]

When the population standard deviation is unknown, the sample size is small and the population is approximately normal, the appropriate test for the mean is

(A) the chi-squared test

(B) the one-sample $z$-test

(C) the one-sample $t$-test

(D) the sign test

**Q.12** [NAT • 2 Marks • Hard]

Consider testing $H_0 : \mu = 100$ against $H_1 : \mu > 100$ with $\sigma = 10$, $n = 25$ and $\alpha = 0.05$. The power of the test when the true mean is $\mu = 104$ is _____ (rounded off to two decimal places). *(Use $z_{0.05} = 1.645$.)*

**Q.13** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) relating the two error types is/are correct?

(A) Power $= 1 - \beta$

(B) $\beta = P(\text{fail to reject } H_0 \mid H_0 \text{ false})$

(C) $\alpha + \beta = 1$ always

(D) $\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$

**Q.14** [NAT • 2 Marks • Moderate]

For a right-tailed $z$-test the observed test statistic is $z = 1.75$. The corresponding $p$-value is _____ (rounded off to three decimal places).

**Q.15** [MCQ • 2 Marks • Hard]

A test performed at $\alpha = 0.05$ yields a $p$-value of 0.03. If the same test were performed at $\alpha = 0.01$,

the decision would be

(A) reject $H_0$

(B) impossible to determine without the sample size

(C) fail to reject $H_0$

(D) unchanged, i.e. reject $H_0$

---

**Q.16** [NAT • 2 Marks • Moderate]

A paired $t$-test is carried out on 10 matched pairs of observations. The number of degrees of freedom of the test statistic is _____.

---

**Q.17** [MCQ • 2 Marks • Moderate]

Two independent samples of size 50 each give $\bar{x}_1 - \bar{x}_2 = 5$, with known $\sigma_1 = \sigma_2 = 10$. The value of the two-sample $z$ statistic is

(A) 5.00 (B) 1.25 (C) 2.50 (D) 0.50

---

**Q.18** [NAT • 2 Marks • Hard]

For the two-sample $z$-test of the previous question, the two-tailed $p$-value is _____ (rounded off to three decimal places).

---

**Q.19** [MSQ • 2 Marks • Hard]

Keeping everything else fixed, which of the following change(s) **increase** the power of a test?

(A) increasing the significance level $\alpha$

(B) increasing the true effect size $|\mu_1 - \mu_0|$

(C) increasing the population standard deviation $\sigma$

(D) increasing the sample size $n$

---

**Q.20** [NAT • 2 Marks • Moderate]

A sample of size 36 gives $\bar{x} = 12.5$. For testing $H_0 : \mu = 12$ with $\sigma = 1.5$, the value of the test statistic is _____ (rounded off to two decimal places).

---

**Q.21** [MCQ • 2 Marks • Moderate]

In the Neyman–Pearson framework of hypothesis testing, the null hypothesis $H_0$

(A) is always the statement of inequality

(B) is the hypothesis the experimenter wishes to prove

(C) is retained unless the data provide sufficient evidence against it

(D) is accepted as proved whenever the test is not significant

---

**Q.22** [NAT • 2 Marks • Hard]

The number of degrees of freedom of a pooled two-sample $t$-test with $n_1 = 12$ and $n_2 = 15$ is _____.

---

## 9. Chi-Squared Test & Descriptive Statistics

**22 questions** • 4 one-mark, 18 two-mark • Suggested time: 60 minutes

**Q.1** [MCQ • 1 Mark • Easy]

In a chi-squared goodness-of-fit test with $k$ categories and no parameters estimated from the data, the number of degrees of freedom is

(A) $k-2$ (B) $k-1$ (C) $n-k$ (D) $k$

**Q.2** [NAT • 1 Mark • Easy]

For a chi-squared test of independence on a $3 \times 4$ contingency table, the number of degrees of freedom is _____.

**Q.3** [MCQ • 1 Mark • Easy]

The median of the data set {3, 7, 2, 9, 4} is

(A) 4 (B) 7 (C) 5 (D) 3

**Q.4** [NAT • 1 Mark • Easy]

For the data $2, 4, 4, 4, 5, 5, 7, 9$, the population standard deviation is _____.

**Q.5** [MCQ • 2 Marks • Moderate]

For the data set $2, 4, 4, 4, 5, 5, 7, 9$, the correct ordering of the three measures of central tendency is

(A) mean = median = mode (B) median < mode < mean (C) mode < median < mean (D) mean < median < mode

**Q.6** [NAT • 2 Marks • Moderate]

A die is rolled 60 times and the observed frequencies of the faces 1 to 6 are $8, 12, 10, 9, 11, 10$. The value of the chi-squared goodness-of-fit statistic for testing whether the die is fair is _____ (rounded off to two decimal places).

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the Pearson chi-squared test is/are correct?

(A) The test is a right-tailed test

(B) It requires the underlying observations to be normally distributed

(C) The expected frequency in each cell should normally be at least 5

(D) It is applied to counts of categorical data

**Q.8** [NAT • 2 Marks • Moderate]

If $\mathrm{Cov}(X, Y) = 12$, $\sigma_X = 4$ and $\sigma_Y = 5$, then the Pearson correlation coefficient is _____ (rounded off to two decimal places).

**Q.9** [MCQ • 2 Marks • Moderate]

A Pearson correlation coefficient of $r = -0.9$ between two variables indicates

(A) a weak negative linear association

(B) a strong positive linear association

(C) no association at all

(D) a strong negative linear association

**Q.10** [NAT • 2 Marks • Moderate]

For the data $10, 20, 30, 40, 50$, the sample variance (with divisor $n - 1$) is _____.

**Q.11** [MSQ • 2 Marks • Moderate]

Which of the following summary measures is/are **resistant** to outliers?

(A) the range

(B) the interquartile range

(C) the arithmetic mean

(D) the median

**Q.12** [NAT • 2 Marks • Moderate]

For the $2 \times 2$ contingency table below, the value of the chi-squared statistic for testing independence (without continuity correction) is _____ (rounded off to two decimal places).

| | $B_1$ | $B_2$ | Total |
|---|---|---|---|
| $A_1$ | 20 | 30 | 50 |
| $A_2$ | 30 | 20 | 50 |
| Total | 50 | 50 | 100 |

**Q.13** [MCQ • 2 Marks • Moderate]

For the contingency table of the previous question, using $\chi^2_{0.05,\,1} = 3.841$, the conclusion at the 5% level of significance is

(A) reject the hypothesis of independence of the two attributes

(B) do not reject the hypothesis of independence

(C) the test cannot be applied because the sample is too small

(D) the two attributes are perfectly correlated

**Q.14** [NAT • 2 Marks • Hard]

For the data $4, 8, 6, 10, 12$, the sample standard deviation (with divisor $n - 1$) is _____ (rounded off to three decimal places).

**Q.15** [MCQ • 2 Marks • Moderate]

Among the mean, median and mode, the measure most strongly affected by extreme observations is

(A) all are equally affected

(B) the mode

(C) the median

(D) the mean

**Q.16** [NAT • 2 Marks • Moderate]

Every observation of a data set with standard deviation 4 is transformed as $y = 3x + 5$. The standard deviation of the transformed data is _____.

**Q.17** [MCQ • 2 Marks • Hard]

For a distribution that is positively (right) skewed, the usual ordering of the measures of central tendency is

(A) median < mean < mode (B) mode < median < mean (C) mean < median < mode (D) mean = median = mode

**Q.18** [NAT • 2 Marks • Moderate]

For a data set with $Q_1 = 15$ and $Q_3 = 35$, the interquartile range is _____.

**Q.19** [MSQ • 2 Marks • Hard]

Which of the following statement(s) about the Pearson correlation coefficient $r$ is/are correct?

(A) $r$ is free of the units of measurement

(B) $-1 \le r \le 1$

(C) $r$ is unchanged if both variables are multiplied by positive constants

(D) $r = 0$ implies that the two variables are independent

**Q.20** [NAT • 2 Marks • Hard]

For the paired data $(1, 2), (2, 4), (3, 5), (4, 4), (5, 5)$, the Pearson correlation coefficient is _____ (rounded off to three decimal places).

**Q.21** [MCQ • 2 Marks • Moderate]

The Pearson chi-squared test statistic is given by

(A) $\sum \dfrac{(O_i - E_i)^2}{E_i}$ (B) $\sum \dfrac{(O_i - E_i)^2}{O_i}$

(C) $\sum (O_i - E_i)^2$ (D) $\sum \dfrac{|O_i - E_i|}{E_i}$

**Q.22** [NAT • 2 Marks • Hard]

A goodness-of-fit test for uniformity over 5 categories is based on 100 observations with observed frequencies $30, 20, 25, 15, 10$. The value of the chi-squared statistic is _____ (rounded off to two decimal places).

## 10. Conditional Expectation, Variance & Covariance

**22 questions** • 4 one-mark, 18 two-mark • Suggested time: 60 minutes

**Q.1** [MCQ • 1 Mark • Easy]

The law of total expectation states that, for random variables $X$ and $Y$,

(A) $E[XY] = E[X]E[Y]$ (B) $E[X] = E\big[E[X \mid Y]\big]$

(C) $E[X] = E[X \mid Y]$ (D) $E[X \mid Y] = E[X]E[Y]$

**Q.2** [NAT • 1 Mark • Easy]

If $\mathrm{Var}(X) = 5$, then $\mathrm{Var}(2X + 3) =$ _____.

**Q.3** [MCQ • 1 Mark • Easy]

For any random variable $X$ with finite variance, $\mathrm{Cov}(X, X)$ equals

(A) $E[X]^2$ (B) 1 (C) 0 (D) $\mathrm{Var}(X)$

**Q.4** [NAT • 1 Mark • Easy]

If $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$ and $\mathrm{Cov}(X, Y) = 6$, then $\mathrm{Var}(X + Y) =$ _____.

**Q.5** [MCQ • 2 Marks • Moderate]

The law of total variance states that

(A) $\mathrm{Var}(X) = \mathrm{Var}\big(E[X \mid Y]\big) - E\big[\mathrm{Var}(X \mid Y)\big]$

(B) $\mathrm{Var}(X) = \mathrm{Var}\big(E[X \mid Y]\big)$

(C) $\mathrm{Var}(X) = E\big[\mathrm{Var}(X \mid Y)\big] + \mathrm{Var}\big(E[X \mid Y]\big)$

(D) $\mathrm{Var}(X) = E\big[\mathrm{Var}(X \mid Y)\big]$

**Q.6** [NAT • 2 Marks • Moderate]

Let $X \sim \mathrm{Uniform}(0, 1)$ and, given $X = x$, let $Y \sim \mathrm{Bernoulli}(x)$. Then $E[Y] =$ _____ (rounded off to two decimal places).

**Q.7** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about covariance is/are correct?

(A) $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$

(B) $\mathrm{Cov}(aX + b,\ cY + d) = ac\,\mathrm{Cov}(X, Y)$

(C) $\mathrm{Cov}(X, Y) = 0$ implies that $X$ and $Y$ are independent

(D) Independence of $X$ and $Y$ implies $\mathrm{Cov}(X, Y) = 0$

**Q.8** [NAT • 2 Marks • Moderate]

If $E[X] = 2$, $E[Y] = 3$ and $E[XY] = 8$, then $\mathrm{Cov}(X, Y) =$ _____.

**Q.9** [MCQ • 2 Marks • Moderate]

If $\mathrm{Var}(X) = 9$, $\mathrm{Var}(Y) = 16$ and the correlation between $X$ and $Y$ is 0.5, then $\mathrm{Var}(X - Y)$ equals

(A) 13 (B) 37 (C) 1 (D) 25

**Q.10** [NAT • 2 Marks • Moderate]

For the joint PDF $f(x, y) = 2$ on $\{0 < x < y < 1\}$, the value of $E[X]$ is _____ (rounded off to three decimal places).

**Q.11** [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the conditional expectation $E[X \mid Y]$ is/are correct?

(A) $E\big[E[X \mid Y]\big] = E[X]$

(B) $E\big[g(Y)X \mid Y\big] = g(Y)\,E[X \mid Y]$

(C) $E[X \mid Y]$ is always a constant

(D) $E[X \mid Y]$ is a random variable which is a function of $Y$

**Q.12** [NAT • 2 Marks • Hard]

A fair six-sided die is rolled once, giving the value $N$. A fair coin is then tossed $N$ times. The expected number of heads obtained is _____ (rounded off to two decimal places).

**Q.13** [MCQ • 2 Marks • Hard]

For the two-stage experiment of the previous question (die then $N$ coin tosses), the variance of the number of heads obtained is closest to

(A) 0.875 (B) 2.917 (C) 1.750 (D) 1.604

**Q.14** [NAT • 2 Marks • Moderate]

Two random variables have correlation 0.8, $\sigma_X = 5$ and $\sigma_Y = 10$. Their covariance is _____.

**Q.15** [MCQ • 2 Marks • Moderate]

If $Y = 3X + 2$ where $X$ has a finite positive variance, then the correlation coefficient between $X$ and $Y$ is

(A) 1 (B) 3 (C) 0 (D) $-1$

**Q.16** [NAT • 2 Marks • Moderate]

If $E[X] = 3$ and $E[X^2] = 13$, then the standard deviation of $X$ is _____.

**Q.17** [MSQ • 2 Marks • Hard]

Which of the following statement(s) about variance is/are correct?

(A) $\mathrm{Var}(X + c) = \mathrm{Var}(X)$ for any constant $c$

(B) $\mathrm{Var}(X) \geq 0$ always

(C) $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$

(D) $\mathrm{Var}(aX) = a\,\mathrm{Var}(X)$

---

**Q.18** [NAT • 2 Marks • Hard]

Let $N \sim \text{Poisson}(5)$ and, given $N = n$, let $X \sim \text{Binomial}(n, 0.4)$. Then $\mathrm{Var}(X) =$ _____ (rounded off to two decimal places).

---

**Q.19** [MCQ • 2 Marks • Moderate]

If $X$ and $Y$ are independent with $\mathrm{Var}(X) = 4$ and $\mathrm{Var}(Y) = 9$, then $\mathrm{Var}(2X - 3Y)$ equals

(A) 97 (B) 25 (C) 49 (D) 13

---

**Q.20** [NAT • 2 Marks • Hard]

Let $X$ be uniform on $\{1, 2, 3\}$ and, given $X = x$, let $Y$ be uniform on $\{1, 2, \ldots, x\}$. Then $E[Y] =$ _____ (rounded off to two decimal places).

---

**Q.21** [MCQ • 2 Marks • Hard]

For the joint PDF $f(x, y) = 4xy$ on the unit square $0 < x < 1,\ 0 < y < 1$, the covariance $\mathrm{Cov}(X, Y)$ equals

(A) 0 (B) $\frac{1}{36}$ (C) $\frac{4}{9}$ (D) $\frac{1}{9}$

---

**Q.22** [NAT • 2 Marks • Hard]

Let $X$ be the number of heads in three tosses of a fair coin and let $Y = X^2$. Then $\mathrm{Cov}(X, Y) =$ _____ (rounded off to two decimal places).

---

# Appendix A — Standard Normal Table ($\Phi(z) = P(Z \le z)$)

| $z$ | 0.00 | 0.01 | 0.02 | 0.03 | 0.04 | 0.05 | 0.06 | 0.07 | 0.08 | 0.09 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.0 | 0.5000 | 0.5040 | 0.5080 | 0.5120 | 0.5160 | 0.5199 | 0.5239 | 0.5279 | 0.5319 | 0.5359 |
| 0.1 | 0.5398 | 0.5438 | 0.5478 | 0.5517 | 0.5557 | 0.5596 | 0.5636 | 0.5675 | 0.5714 | 0.5753 |
| 0.2 | 0.5793 | 0.5832 | 0.5871 | 0.5910 | 0.5948 | 0.5987 | 0.6026 | 0.6064 | 0.6103 | 0.6141 |
| 0.3 | 0.6179 | 0.6217 | 0.6255 | 0.6293 | 0.6331 | 0.6368 | 0.6406 | 0.6443 | 0.6480 | 0.6517 |
| 0.4 | 0.6554 | 0.6591 | 0.6628 | 0.6664 | 0.6700 | 0.6736 | 0.6772 | 0.6808 | 0.6844 | 0.6879 |
| 0.5 | 0.6915 | 0.6950 | 0.6985 | 0.7019 | 0.7054 | 0.7088 | 0.7123 | 0.7157 | 0.7190 | 0.7224 |
| 0.6 | 0.7257 | 0.7291 | 0.7324 | 0.7357 | 0.7389 | 0.7422 | 0.7454 | 0.7486 | 0.7517 | 0.7549 |
| 0.7 | 0.7580 | 0.7611 | 0.7642 | 0.7673 | 0.7704 | 0.7734 | 0.7764 | 0.7794 | 0.7823 | 0.7852 |
| 0.8 | 0.7881 | 0.7910 | 0.7939 | 0.7967 | 0.7995 | 0.8023 | 0.8051 | 0.8078 | 0.8106 | 0.8133 |
| 0.9 | 0.8159 | 0.8186 | 0.8212 | 0.8238 | 0.8264 | 0.8289 | 0.8315 | 0.8340 | 0.8365 | 0.8389 |
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554 | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770 | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962 | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | 0.9131 | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279 | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406 | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515 | 0.9525 | 0.9535 | 0.9545 |
| 1.7 | 0.9554 | 0.9564 | 0.9573 | 0.9582 | 0.9591 | 0.9599 | 0.9608 | 0.9616 | 0.9625 | 0.9633 |
| 1.8 | 0.9641 | 0.9649 | 0.9656 | 0.9664 | 0.9671 | 0.9678 | 0.9686 | 0.9693 | 0.9699 | 0.9706 |
| 1.9 | 0.9713 | 0.9719 | 0.9726 | 0.9732 | 0.9738 | 0.9744 | 0.9750 | 0.9756 | 0.9761 | 0.9767 |
| 2.0 | 0.9772 | 0.9778 | 0.9783 | 0.9788 | 0.9793 | 0.9798 | 0.9803 | 0.9808 | 0.9812 | 0.9817 |
| 2.1 | 0.9821 | 0.9826 | 0.9830 | 0.9834 | 0.9838 | 0.9842 | 0.9846 | 0.9850 | 0.9854 | 0.9857 |
| 2.2 | 0.9861 | 0.9864 | 0.9868 | 0.9871 | 0.9875 | 0.9878 | 0.9881 | 0.9884 | 0.9887 | 0.9890 |
| 2.3 | 0.9893 | 0.9896 | 0.9898 | 0.9901 | 0.9904 | 0.9906 | 0.9909 | 0.9911 | 0.9913 | 0.9916 |
| 2.4 | 0.9918 | 0.9920 | 0.9922 | 0.9925 | 0.9927 | 0.9929 | 0.9931 | 0.9932 | 0.9934 | 0.9936 |
| 2.5 | 0.9938 | 0.9940 | 0.9941 | 0.9943 | 0.9945 | 0.9946 | 0.9948 | 0.9949 | 0.9951 | 0.9952 |
| 2.6 | 0.9953 | 0.9955 | 0.9956 | 0.9957 | 0.9959 | 0.9960 | 0.9961 | 0.9962 | 0.9963 | 0.9964 |
| 2.7 | 0.9965 | 0.9966 | 0.9967 | 0.9968 | 0.9969 | 0.9970 | 0.9971 | 0.9972 | 0.9973 | 0.9974 |
| 2.8 | 0.9974 | 0.9975 | 0.9976 | 0.9977 | 0.9977 | 0.9978 | 0.9979 | 0.9979 | 0.9980 | 0.9981 |
| 2.9 | 0.9981 | 0.9982 | 0.9982 | 0.9983 | 0.9984 | 0.9984 | 0.9985 | 0.9985 | 0.9986 | 0.9986 |
| 3.0 | 0.9987 | 0.9987 | 0.9987 | 0.9988 | 0.9988 | 0.9989 | 0.9989 | 0.9989 | 0.9990 | 0.9990 |
| 3.1 | 0.9990 | 0.9991 | 0.9991 | 0.9991 | 0.9992 | 0.9992 | 0.9992 | 0.9992 | 0.9993 | 0.9993 |
| 3.2 | 0.9993 | 0.9993 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9994 | 0.9995 | 0.9995 | 0.9995 |
| 3.3 | 0.9995 | 0.9995 | 0.9995 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9996 | 0.9997 |
| 3.4 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9997 | 0.9998 |

For $z < 0$ use $\Phi(-z) = 1 - \Phi(z)$. Common upper-tail points: $z_{0.10} = 1.282$, $z_{0.05} = 1.645$, $z_{0.025} = 1.960$, $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

## Appendix B — $t$-Distribution: upper-tail critical values $t_{\alpha,\nu}$

| $\nu$ | 0.1 | 0.05 | 0.025 | 0.01 | 0.005 |
|---|---|---|---|---|---|
| 1 | 3.078 | 6.314 | 12.706 | 31.821 | 63.657 |
| 2 | 1.886 | 2.920 | 4.303 | 6.965 | 9.925 |
| 3 | 1.638 | 2.353 | 3.182 | 4.541 | 5.841 |
| 4 | 1.533 | 2.132 | 2.776 | 3.747 | 4.604 |
| 5 | 1.476 | 2.015 | 2.571 | 3.365 | 4.032 |
| 6 | 1.440 | 1.943 | 2.447 | 3.143 | 3.707 |
| 7 | 1.415 | 1.895 | 2.365 | 2.998 | 3.499 |
| 8 | 1.397 | 1.860 | 2.306 | 2.896 | 3.355 |
| 9 | 1.383 | 1.833 | 2.262 | 2.821 | 3.250 |
| 10 | 1.372 | 1.812 | 2.228 | 2.764 | 3.169 |
| 11 | 1.363 | 1.796 | 2.201 | 2.718 | 3.106 |
| 12 | 1.356 | 1.782 | 2.179 | 2.681 | 3.055 |
| 13 | 1.350 | 1.771 | 2.160 | 2.650 | 3.012 |
| 14 | 1.345 | 1.761 | 2.145 | 2.624 | 2.977 |
| 15 | 1.341 | 1.753 | 2.131 | 2.602 | 2.947 |
| 16 | 1.337 | 1.746 | 2.120 | 2.583 | 2.921 |
| 17 | 1.333 | 1.740 | 2.110 | 2.567 | 2.898 |
| 18 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 |
| 19 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 |
| 20 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 |
| 21 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 |
| 22 | 1.321 | 1.717 | 2.074 | 2.508 | 2.819 |
| 23 | 1.319 | 1.714 | 2.069 | 2.500 | 2.807 |
| 24 | 1.318 | 1.711 | 2.064 | 2.492 | 2.797 |
| 25 | 1.316 | 1.708 | 2.060 | 2.485 | 2.787 |
| 26 | 1.315 | 1.706 | 2.056 | 2.479 | 2.779 |
| 27 | 1.314 | 1.703 | 2.052 | 2.473 | 2.771 |
| 28 | 1.313 | 1.701 | 2.048 | 2.467 | 2.763 |
| 29 | 1.311 | 1.699 | 2.045 | 2.462 | 2.756 |
| 30 | 1.310 | 1.697 | 2.042 | 2.457 | 2.750 |
| 40 | 1.303 | 1.684 | 2.021 | 2.423 | 2.704 |
| 60 | 1.296 | 1.671 | 2.000 | 2.390 | 2.660 |
| 120 | 1.289 | 1.658 | 1.980 | 2.358 | 2.617 |
| $\infty$ | 1.282 | 1.645 | 1.960 | 2.326 | 2.576 |

## Appendix C — $\chi^2$ Distribution: upper-tail critical values $\chi^2_{\alpha,\nu}$

| $\nu$ | 0.995 | 0.975 | 0.95 | 0.1 | 0.05 | 0.025 | 0.01 |
|---|---|---|---|---|---|---|---|
| 1 | 0.000 | 0.001 | 0.004 | 2.706 | 3.841 | 5.024 | 6.635 |
| 2 | 0.010 | 0.051 | 0.103 | 4.605 | 5.991 | 7.378 | 9.210 |
| 3 | 0.072 | 0.216 | 0.352 | 6.251 | 7.815 | 9.348 | 11.345 |
| 4 | 0.207 | 0.484 | 0.711 | 7.779 | 9.488 | 11.143 | 13.277 |
| 5 | 0.412 | 0.831 | 1.145 | 9.236 | 11.070 | 12.833 | 15.086 |
| 6 | 0.676 | 1.237 | 1.635 | 10.645 | 12.592 | 14.449 | 16.812 |
| 7 | 0.989 | 1.690 | 2.167 | 12.017 | 14.067 | 16.013 | 18.475 |
| 8 | 1.344 | 2.180 | 2.733 | 13.362 | 15.507 | 17.535 | 20.090 |
| 9 | 1.735 | 2.700 | 3.325 | 14.684 | 16.919 | 19.023 | 21.666 |
| 10 | 2.156 | 3.247 | 3.940 | 15.987 | 18.307 | 20.483 | 23.209 |
| 11 | 2.603 | 3.816 | 4.575 | 17.275 | 19.675 | 21.920 | 24.725 |
| 12 | 3.074 | 4.404 | 5.226 | 18.549 | 21.026 | 23.337 | 26.217 |
| 13 | 3.565 | 5.009 | 5.892 | 19.812 | 22.362 | 24.736 | 27.688 |
| 14 | 4.075 | 5.629 | 6.571 | 21.064 | 23.685 | 26.119 | 29.141 |
| 15 | 4.601 | 6.262 | 7.261 | 22.307 | 24.996 | 27.488 | 30.578 |
| 16 | 5.142 | 6.908 | 7.962 | 23.542 | 26.296 | 28.845 | 32.000 |
| 17 | 5.697 | 7.564 | 8.672 | 24.769 | 27.587 | 30.191 | 33.409 |
| 18 | 6.265 | 8.231 | 9.390 | 25.989 | 28.869 | 31.526 | 34.805 |
| 19 | 6.844 | 8.907 | 10.117 | 27.204 | 30.144 | 32.852 | 36.191 |
| 20 | 7.434 | 9.591 | 10.851 | 28.412 | 31.410 | 34.170 | 37.566 |
| 21 | 8.034 | 10.283 | 11.591 | 29.615 | 32.671 | 35.479 | 38.932 |
| 22 | 8.643 | 10.982 | 12.338 | 30.813 | 33.924 | 36.781 | 40.289 |
| 23 | 9.260 | 11.689 | 13.091 | 32.007 | 35.172 | 38.076 | 41.638 |
| 24 | 9.886 | 12.401 | 13.848 | 33.196 | 36.415 | 39.364 | 42.980 |
| 25 | 10.520 | 13.120 | 14.611 | 34.382 | 37.652 | 40.646 | 44.314 |
| 26 | 11.160 | 13.844 | 15.379 | 35.563 | 38.885 | 41.923 | 45.642 |
| 27 | 11.808 | 14.573 | 16.151 | 36.741 | 40.113 | 43.195 | 46.963 |
| 28 | 12.461 | 15.308 | 16.928 | 37.916 | 41.337 | 44.461 | 48.278 |
| 29 | 13.121 | 16.047 | 17.708 | 39.087 | 42.557 | 45.722 | 49.588 |
| 30 | 13.787 | 16.791 | 18.493 | 40.256 | 43.773 | 46.979 | 50.892 |

# Answer Key

Answers only. For NAT questions any value within the usual GATE tolerance of the stated figure is correct.

## 1. Counting & Basic Probability

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | C | **2** | 50400 | **3** | A | **4** | 0.139 | **5** | B |
| **6** | 84 | **7** | A, B, C | **8** | 120 | **9** | D | **10** | 150 |
| **11** | C | **12** | 231 | **13** | A, B, D | **14** | 0.286 | **15** | D |
| **16** | 0.556 | **17** | C | **18** | 103776 | **19** | D | **20** | 0.482 |
| **21** | B, C, D | **22** | 2880 | **23** | D | **24** | 0.444 | | |

## 2. Conditional & Joint Probability

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | B | **2** | 0.333 | **3** | B | **4** | 0.12 | **5** | 0.286 |
| **6** | 1.05 | **7** | A, B, D | **8** | D | **9** | 0.032 | **10** | C |
| **11** | 0.0625 | **12** | C | **13** | 0.0129 | **14** | B | **15** | 0.133 |
| **16** | A, C, D | **17** | 0.54 | **18** | B | **19** | 0.75 | **20** | D |
| **21** | 0.775 | **22** | A, B | **23** | 0.143 | **24** | A | | |

## 3. Bayes' Theorem

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | B | **2** | 0.167 | **3** | B | **4** | 0.625 | **5** | A |
| **6** | 0.037 | **7** | B, C, D | **8** | 0.50 | **9** | B | **10** | 0.870 |
| **11** | B | **12** | 0.667 | **13** | A | **14** | 0.517 | **15** | A, B, D |
| **16** | 0.414 | **17** | D | **18** | 0.652 | **19** | C | **20** | 0.857 |
| **21** | A, B, C | **22** | 0.45 | | | | | | |

## 4. Discrete Random Variables & PMF

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | B | **2** | 0.135 | **3** | C | **4** | 0.346 | **5** | D |
| **6** | 0.801 | **7** | A, B, D | **8** | 6 | **9** | A | **10** | 0.195 |
| **11** | C | **12** | 3 | **13** | A, B, C | **14** | 20 | **15** | C |
| **16** | 0.273 | **17** | C | **18** | 6 | **19** | D | **20** | 2.25 |
| **21** | A, B, D | **22** | 0.736 | **23** | D | **24** | 0.034 | **25** | B |
| **26** | 3.667 | | | | | | | | |

## 5. Continuous Random Variables & PDF/CDF

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | C | **2** | 0.135 | **3** | A | **4** | 0.933 | **5** | B |
| **6** | 0.819 | **7** | B, C, D | **8** | 0.111 | **9** | D | **10** | 2.079 |
| **11** | B | **12** | 0.023 | **13** | A | **14** | 10 | **15** | A, B, D |
| **16** | 0.50 | **17** | A | **18** | 0.25 | **19** | D | **20** | 0.05 |
| **21** | B, C, D | **22** | 0.393 | **23** | C | **24** | 26.58 | **25** | B |
| **26** | 1.00 | | | | | | | | |

## 6. $t$-distribution & Chi-Squared Distribution

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | C | **2** | 16 | **3** | C | **4** | 1.25 | **5** | B |
| **6** | 15 | **7** | A, B, D | **8** | 1.782 | **9** | B | **10** | 18.307 |
| **11** | A, B, C | **12** | 34.56 | **13** | D | **14** | 15 | **15** | B |
| **16** | 2.086 | **17** | D | **18** | 7 | **19** | A | **20** | 0.05 |

## 7. Central Limit Theorem & Confidence Intervals

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | C | **2** | 2 | **3** | C | **4** | 3.92 | **5** | B |
| **6** | 97 | **7** | B, C, D | **8** | 4.128 | **9** | D | **10** | 93.56 |
| **11** | D | **12** | 0.042 | **13** | B, C, D | **14** | 0.023 | **15** | C |
| **16** | 0.954 | **17** | D | **18** | 0.159 | **19** | A, C, D | **20** | 38.29 |
| **21** | D | **22** | 246 | | | | | | |

## 8. Hypothesis Testing: $z$-test & $t$-test

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | A | **2** | 1.96 | **3** | B | **4** | 0.80 | **5** | C |
| **6** | 2.00 | **7** | A, B, C | **8** | 2.50 | **9** | B | **10** | 2.861 |
| **11** | C | **12** | 0.64 | **13** | A, B, D | **14** | 0.040 | **15** | C |
| **16** | 9 | **17** | C | **18** | 0.012 | **19** | A, B, D | **20** | 2.00 |
| **21** | C | **22** | 25 | | | | | | |

## 9. Chi-Squared Test & Descriptive Statistics

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | B | **2** | 6 | **3** | A | **4** | 2 | **5** | C |
| **6** | 1.00 | **7** | A, C, D | **8** | 0.60 | **9** | D | **10** | 250 |
| **11** | B, D | **12** | 4.00 | **13** | A | **14** | 3.162 | **15** | D |
| **16** | 12 | **17** | B | **18** | 20 | **19** | A, B, C | **20** | 0.775 |
| **21** | A | **22** | 12.50 | | | | | | |

## 10. Conditional Expectation, Variance & Covariance

| Q | Ans | Q | Ans | Q | Ans | Q | Ans | Q | Ans |
|---|---|---|---|---|---|---|---|---|---|
| **1** | B | **2** | 20 | **3** | D | **4** | 25 | **5** | C |
| **6** | 0.50 | **7** | A, B, D | **8** | 2 | **9** | A | **10** | 0.333 |
| **11** | A, B, D | **12** | 1.75 | **13** | D | **14** | 40 | **15** | A |
| **16** | 2 | **17** | A, B, C | **18** | 2.00 | **19** | A | **20** | 1.50 |
| **21** | A | **22** | 2.25 | | | | | | |